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EEG Signals Recognition of Active Brain-Computer Interface Based on Sequential Encoding

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***Abstract*—To complement the existing experimental paradigm of motor imagery, an active brain-computer interface (BCI) experimental paradigm based on the sequential encoding of speech imagery and motor imagery is proposed. At the same time, a sub-time window filter bank common spatial pattern (STWFBCSP) algorithm is proposed to overcome the deficiency of common spatial pattern (CSP). In this algorithm, the imagery period is divided into five overlapping time windows and each time window is filtered by multi-frequency. Feature extraction and selection are achieved by CSP and mutual information, respectively. The final decision output is determined by the voting results of five support vector machines (SVMs). Using STWFBCSP, the average classification accuracy of electroencephalogram (EEG) signals of 12 subjects is 84.87%. Compared with CSP, filter bank common spatial pattern (FBCSP), and common time-frequency-spatial pattern (CTFSP) algorithm, the result of the proposed algorithm is improved by 14.38%, 9.02%, and 3.97%, respectively. The results show that the EEG signals of this experimental paradigm can be better extracted and classified by STWFBCSP, and the practicability of BCI is also improved.**

Keywords-electroencephalogram; sequential encoding; common spatial pattern; support vector machine

I. INTRODUCTION

Brain-computer interface (BCI) is a new human-computer interaction system, which establishes a new information exchange and control channel between humans and the surrounding environment [1]. In view of the problems faced in the application of BCI, there are two main directions of innovation at present: on the one hand, innovation of experimental paradigm, on the other hand, innovation of algorithm [2]. Due to the constraints of spatial resolution of electroencephalogram (EEG) signals, motor imagery has few operational dimensions, mainly including hands, feet, and tongue. The limitation of spatial resolution can be broken by using the high temporal resolution of EEG signals. Four types of sequential complex limb motor imagery are proposed by Yi et al [3]. In addition, to realize the four control modes of the wheelchairs, a sequence paradigm is proposed with left- and right-hand motor imagery tasks [4]. As there are few experimental paradigms based on sequential encoding, we propose an active BCI experimental paradigm based on the sequential encoding of speech imagery and motor imagery.

In the BCI system, common spatial pattern (CSP) is very prevalent and efficient [5]. The effectiveness of CSP depends on the selected frequency band. Different subjects have different optimal frequency bands. Therefore, it is necessary to optimize the filtering frequency band to improve the performance of CSP. A filter bank common spatial pattern (FBCSP) is proposed, which uses multi-band filtering to improve the performance of the CSP [6]. A local region filter bank common spatial pattern (LRFBCSP) is proposed. It discusses the local CSP features generated by a single channel and the use of filter banks to optimize local CSP features [7].

Apart from the improvement of the filtering frequency band, another very significant but rarely studied problem is to choose a proper time window to segment the EEG signals. It can make the extracted features have strong distinguishability. To optimize the filtering frequency band and time window within the CSP, A temporally constrained sparse group spatial pattern (TSGSP) is proposed [8]. In order to effectively extract the time, frequency, and spatial features of EEG signals, a common time-frequency-spatial pattern (CTFSP) is proposed [9]. In view of the poor effect of previous algorithms in dealing with EEG signals in this experimental paradigm, a sub-time window filter bank common spatial pattern (STWFBCSP) algorithm is proposed. It is an optimization of CTFSP in the time window. In the end, the effectiveness of STWFBCSP is verified by our collected EEG signals.

II. EXPERIMENTAL DESIGN

A. Experimental Paradigm

The experimental subjects in this paper are 12 young subjects (8 males, 4 females, aged between 22 and 26 years old) who are healthy and in good psychological condition. 12 subjects are represented by S1-S12 in turn. 12 subjects ensure sufficient rest before the experiment. They are forbidden to drink tea, alcohol, and coffee for 24 hours before the experiment [10]. because they stimulate the brain. The

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experiment has been approved by the Ethics Review Committee of Guangzhou University. Before the experiment, 12 subjects read the descriptions and precautions related to the experiment, and signed the "Experiment Informed Consent".

The sequence diagram of an experiment is shown in Fig. 1. An experiment lasts 8 s, and the process of one experiment is as follows: 0-2 s, a "*" appears on the screen, indicating the relax period, and the subjects can take appropriate rest; 2-3 s, the screen displays a "+", it is the ready period, requiring the subjects to prepare well; 3-4 s, two kinds of cue ("→" and "right→") appear randomly, indicating the cue period, telling the subjects what to imagine; 4-8 s, the display screen shows a black screen, indicating the imagery period. If the cue is "→", the subjects are demanded to continuously imagine the right hand flapping up and down. If the cue is " right→", the subjects are asked to read it silently three times quickly in Chinese, and then continuously imagine the right hand flapping up and down. In every run, each cue randomly appears 15 times. Each subject collects a total of 5 runs. Finally, a total of 75 sets of data are collected for each type of imagery.

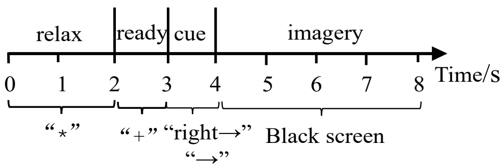


Fig. 1. The sequence diagram of an experiment of sequential encoding

B. Data Acquisition

Imagining the movement of the right hand belongs to motor imagery, which is related to the motor cortex of the brain. Reading Chinese characters silently belongs to speech imagery, which is related to the language cortex of the brain. Therefore, to record EEG signals comprehensively, the 16 electrodes on the cap need to cover the parietal lobe, Wernicke's area, Broca's area, and primary motor cortex. These electrodes are set up according to the international 10-20 system. The position distribution of the 16 channels on the electrode cap is shown in Fig. 2. The sampling frequency of EEG signals is 250Hz.

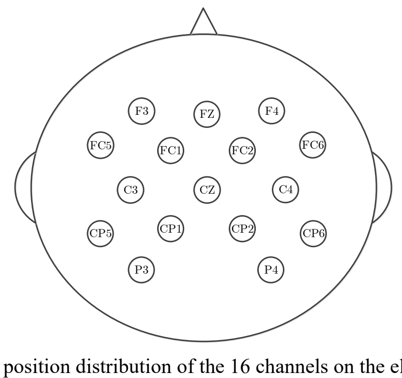


Fig. 2. The position distribution of the 16 channels on the electrode cap according to the international 10-20 system.

III. Methods

A. Preprocessing

First, The EEG signals are filtered by a 6-order Butterworth bandpass filter with a range of 8-30 Hz. Then 50Hz power-frequency interference is removed. In the end, the EEG signals after the above processing are divided into 7 sub-bands: α band (8-13 Hz), β band (13-30 Hz), and other five sub-bands (8-10 Hz, 10-13 Hz, 13-18 Hz, 18-23 Hz and 23-30 Hz).

B. Feature Extraction

Spatial features of two different kinds of imagery tasks can be extracted by CSP. The spatial components of each class can be extracted from multi-channel EEG signals by the algorithm. An optimal projection matrix W is calculated by CSP [11]. It is assumed that the EEG acquisition device has M channels and each channel collects N data points in each experiment. each experiment can be regarded as a matrix E of $M \times N$. Then the eigenmatrix Z is

$$Z = WE \tag{1}$$

The EEG signals are filtered by a spatial filter to obtain the matrix Z_j (j = 1, 2, ... 2m). The eigenvectors of single experimental data are calculated as follows

$$V_j = \ln\left(\frac{\operatorname{var}(Z_j)}{\sum_{i=1}^{2m}\operatorname{var}(Z_i)}\right) \tag{2}$$

In Formula (2), the value of m is 2, and var (•) is the calculation function of variance.

C. Feature Selection

The feature selection algorithm based on mutual information, this problem is defined as, assumes that a set F contains d features, find a subset S with b features ($b < d$), so as to maximize the mutual information I (S; Ω) [12]. In pattern classification problems, classes are discrete and input features are generally continuous. Therefore, the mutual information between class Ω and input feature X is

$$I(X;\Omega) = H(\Omega) - H(\Omega \mid X) \tag{3}$$

$\omega \in \Omega$, where $\Omega = \{1, ..., N_\omega\}$, N_ω is the total number of classes and the conditional entropy is

$$H(\Omega \mid X) = -\int_X \sum_{\omega=1}^{N_\omega} p(\omega \mid x)\log_2 p(\omega \mid x)\,dx \tag{4}$$

In this paper, the mutual information based best individual feature (MIBIF) algorithm is adopted as the feature selection. In MIBIF, a parameter k needs to be determined to select the k features with the largest mutual information [6]. In this paper, the k value of each subject is different, and the determination of the k value needs to be obtained through multiple debugging.

D.Feature Classification

SVM is a machine learning method, which performs well in solving classification and prediction problems and is widely used [13]. The key to solving SVM is to solve the classification hyperplane. The training dataset can be correctly segmented by the hyperplane and has the largest geometric interval. The hyperplane is

$$w^T x + b = 0 \tag{5}$$

For the nonlinearity of EEG signals, a kernel function satisfying the Mercer condition needs to be set. In this paper, we select the radial basis function $K(x_i, x_c)$ with unlimited mapping dimension as the kernel function. The function is

$$K\left(x_i, x_c\right) = \exp\left(-g \,\|\, x_i - x_c \,\|^2\right) \tag{6}$$

x_c is the center of the kernel function, x_i is the feature vector of training data.

E. ERD/ERS Analyses

Pfurtscheller et al. [14] propose an event-related synchronization/desynchronization (ERS/ERD) analysis method. Assume that A is the energy of EEG signals when performing imagery task, and R is the energy of the EEG signals one second before performing imagery tasks. The following formula can calculate the relative change in energy.

$$D = \frac{A - R}{R} \times 100\% \tag{7}$$

F. Proposed Model

Since speech imagery and motor imagery are performed sequentially, the experimental paradigm designed by us has obvious distinguishing features in time. CSP algorithm extracts spatial features of EEG signals without carrying any temporal features. Therefore, it is necessary to extract temporal features of EEG signals to improve classification performance. In the existing EEG signals processing methods, most of them deal with the data of 0.5-2.5 s in the imagery period of 0-4 s. However, different subjects have different temporal patterns when performing different imagery tasks. Therefore, in order to capture the temporal patterns of the active BCI paradigm based on the sequential encoding of speech imagery and motor imagery, we propose to divide the imagery period into five stages: response stage, imagery stage 1, switching stage, imagery stage 2, and completion stage. They are 4-6 s, 4.5-6.5 s, 5-7 s, 5.5-7.5 s, and 6-8 s respectively.

Different subjects have different speeds and ways of imagery. Therefore, it is a complex problem to check when the subject starts to imagine, how long the speech imagery lasts, when switching, how long the motor imagery lasts, and when it ends [15]. However, to distinguish different imagery tasks, time information of EEG signals needs to be extracted. Therefore, we divided the imagery period into five overlapping time windows of 2 s length.

In order to achieve a better classification performance, The STWFBCSP algorithm is proposed. The structural block diagram of STWFBCSP is shown in Fig. 3. Firstly, the whole imagery period is divided into five overlapping time windows, and multi-band filtering is performed for each time window. Then the multi-band filtered signals are extracted by CSP and selected by MIBIF respectively. The voting results of the five SVMs determine the final output.

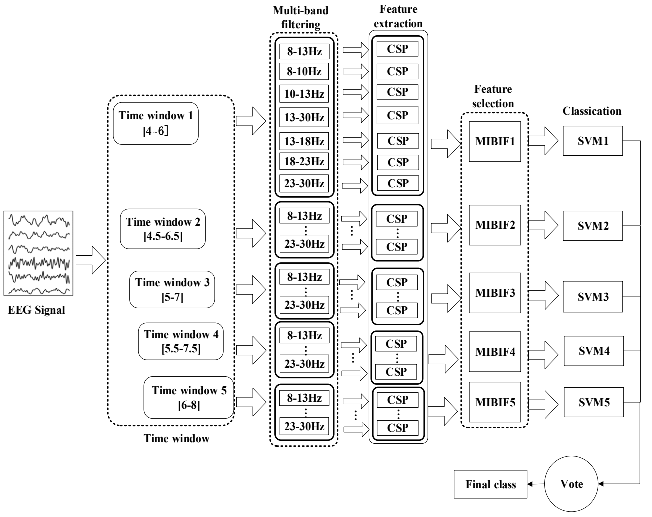
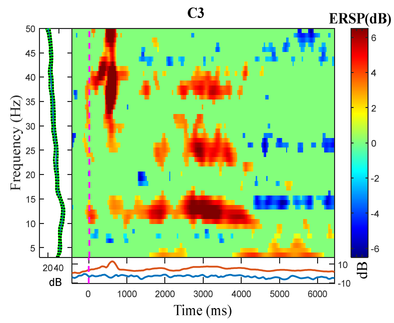


Fig. 3. Block diagram of STWFBCSP

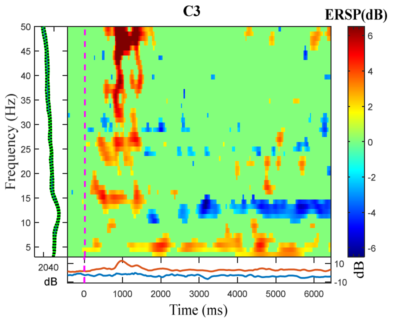
IV. RESULTS & DISCUSSION

A. Analysis

To understand the time-frequency characteristics of EEG signals, we use EEGLAB to plot the event-related spectral perturbation (ERSP) diagram of EEG signals after 4-45 Hz band-pass filtering. The ERSP diagram of channel C3 for subject S3 is shown in Fig. 4. The position of 1 s on the time



(a)



(b)

Fig. 4. The ERSP diagrams of channel C3 for subject S3: (a) speech imagery + motor imagery; (b) motor imagery.

axis is the moment when the imagery starts. As can be seen from Fig. 4, When performing complex imagery, subject S3 performs speech imagery first and then motor imagery. In the first 2 s of the imagery period, the energy of channel C3 is high and the ERS phenomenon appears. In the last 2 s of the imagery period, the energy of channel C3 is low and the ERD phenomenon appears. The energy of channel C3 is always at a lower level when subject S3 only engages in motor imagery. According to the results of ERSP, the experimental paradigm we designed based on sequential encoding has obvious distinguishing characteristics in time.

In order to further compare the difference between the two imageries, The ERD/ERS diagram of channel C3 of subject S3 is drawn in Fig. 5. In terms of the energy of EEG signals, there is a significant difference between motor imagery and speech imagery. When subjects engage in speech imagery and motor imagery, the energy of channel C3 increases and decreases respectively. As can be seen from Fig. 5, when the cue is "right →", the subjects first carry out speech imagery and then perform motor imagery. The ERS phenomenon appears in the first 2 s of the imagery period, and the ERD phenomenon appears in the last 2 s of the imagery period. When subjects only engage in motor imagery, only ERD appears in the whole imagery period. The difference between the two kinds of imageries is mainly reflected in the first two seconds of the imagery period. The results of ERD/ERS further show that the experimental paradigm designed in this paper has significant differences in time.

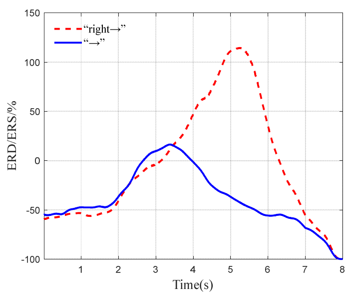
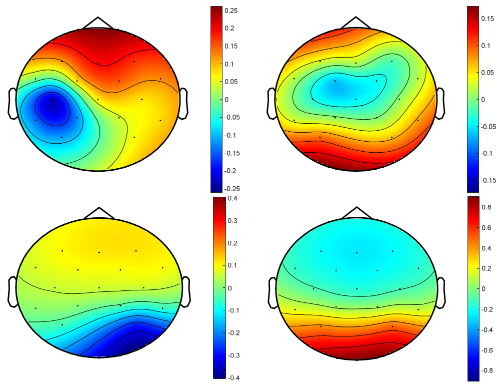


Fig. 5. The ERD/ERS diagram of channel C3 of subject S3.

To verify that this experimental paradigm is time-varying in spatial features, we compare the spatial features of EEG signals on a time axis. According to the difference in time of this experimental paradigm, we draw two pairs of EEG topographic maps. They come from the first two seconds and the last two seconds of the imagery period. The topographic map of subject S3 is shown in Fig. 6. These figures are from the first and last columns of the inverse matrix W^{-1} of the constructed spatial filter. In Fig. 6., the two subgraphs in the first column are from the data of the first two seconds of the imagery period. They correspond to motor imagery and speech imagery respectively. When subject S3 performs motor imagery, the ERD phenomenon appears in the parietal lobe of the left brain. When subject S3 performs speech imagery, ERS phenomenon appears in the frontal lobe. The two subgraphs in the second column are from the data of the last two seconds of the imagery period. Both of them are motor imagery. They come from the same imagery task, and there is not much difference between the two maps.



(a) (b)

Fig. 6. The topographic map of subject S3: (a) the maps of the first two seconds of imagery period for "right →" vs "→"; (b) the maps of the last two seconds of imagery period for "right →" vs "→".

B. Results

The classification results for 12 subjects are shown in Table 1. The classification results of each subject are obtained by averaging the results of cross-validation of 10-fold. Paired sample t-test is used to analyze the significant difference between the results. Comparing the classification accuracy of STWFBCSP proposed by us with CSP, FBCSP, and CTFSP, the following conclusions can be drawn from Table 1. From the average results of 12 subjects, the average classification result obtained by CSP is 70.49%. The average classification result of FBCSP is 75.85%, which is 5.36% higher than that of CSP ($p < 0.01$, paired sample t-test). The average classification result of CTFSP is 80.9%. Compared with CSP and FBCSP, its average classification accuracy is improved by 10.41% ($p < 0.01$) and 5.05% ($p < 0.01$) respectively. The average classification accuracy of STWFBCSP is 84.87%. Compared with CSP, FBCSP, and CTFSP, its average classification accuracy is

Table 1 The classification results of four algorithms for 12 subjects (the highest classification results of each subject is highlighted in bold)

Subject	Methods			
	CSP/%	FBCSP/%	CTFSP/%	STWFBCSP/%
S1	75.00±1.12	73.05±1.55	78.47±2.16	**83.09±0.87**

S2	72.90±1.28	83.46±1.48	85.82±1.52	**87.30±1.14**
S3	89.55±1.48	89.58±1.15	92.07±0.65	**96.10±0.66**
S4	69.12±2.09	75.77±1.10	77.12±1.39	**85.88±1.92**
S5	67.35±1.37	70.41±1.06	75.86±1.75	**80.66±1.17**
S6	62.81±2.19	67.65±1.34	73.54±2.28	**79.52±1.08**
S7	70.36±1.72	74.83±1.33	81.95±2.06	**84.97±1.37**
S8	60.28±2.99	75.59±1.27	75.08±1.76	**80.07±1.09**
S9	75.94±1.16	75.90±0.98	**81.40±1.34**	80.16±1.33
S10	71.59±1.13	76.12±1.68	86.06±1.06	**90.31±0.93**
S11	69.30±1.47	78.63±1.16	80.64±1.47	**88.81±1.40**
S12	61.65±1.61	69.23±1.98	**82.83±1.58**	81.57±1.21
Mean	70.49	75.85	80.90	**84.87**

improved by 14.38% ($p < 0.01$), 9.02% ($p < 0.01$), and 3.97% ($p < 0.01$) respectively. From the individual results of each subject, except for subjects S9 and S12, the classification result of STWFBCSP is better than other algorithms in the other 10 subjects. For STWFBCSP, the classification results of the two subjects are above 90%. Among them, the classification accuracy of S8 is the highest, reaching 96.1%. Meanwhile, the classification results of adding a time window are better than that of not adding a time window.

C. Discussion

Due to the low spatial resolution of EEG signals, the number of instructions of active BCI is relatively small. Taking advantage of the high temporal resolution of EEG signals, our proposed experimental paradigm can overcome this limitation. This paper is the first exploration of the experimental paradigm based on the sequential encoding of speech imagery and motor imagery. Its feasibility can be seen from the time-frequency analysis and classification results. It lays the foundation for the subsequent design of different sequential encoding experimental paradigms. This experimental paradigm has obvious temporal characteristics. The STWFBCSP designed in this paper can not only extract the spatial features of EEG signals in different frequency bands but also extract the features from the perspective of time. Therefore, it obtains higher classification accuracy than CSP and FBCSP.

The classification results of CSP are poor. The main reason is that the features extracted by CSP are only spatial features and lack temporal and frequency features. Our proposed experimental paradigm has obvious distinguishable characteristics in time. The effect of CSP depends on the frequency band of the EEG signals. Therefore, we propose the STWFBCSP algorithm. It can select the optimal feature among the spatial features extracted from multiple frequency bands in each time window. It shows that the features extracted by this method include temporal, frequency, and spatial features. Therefore, it works better than CSP and FBCSP.

For most subjects, the classification results of STWFBCSP are better than CTFSP. It shows that the time window of STWFBCSP is better than CTFSP for this experimental paradigm. The main reason is that STWFBCSP divides the imagery period more carefully, and does not contain a lot of redundant information.

V. CONCLUSION

To increase the number of instructions, we propose an active BCI experimental paradigm based on sequential encoding of speech imagery and motor imagery. It enriches the BCI system of the hybrid model. At the same time, a STWFBCSP algorithm is proposed. It can extract spatial features of different frequency bands in each time window effectively. The STWFBCSP is applied to the EEG signals of this experimental paradigm. The results show that STWFBCSP is superior to other algorithms. From the process of analysis and the results of classification, we can get the feasibility of this experimental paradigm and the effectiveness of STWFBCSP. Therefore, the research of this paper provides a new idea for the application of active BCI based on sequential encoding. In future work, we will propose a multi-classification experimental paradigm based on sequential encoding and propose a better algorithm for multi-classification.

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